Feedback & motor skill learning

What helps a student improve their next attempt?

Effects of Feedback on Students' Motor Skill Learning in Physical Education: A Systematic Review

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IJERPH, 2021 · 23 studies reviewed

School PE · Feedback design · Implications for XR

Practice gives students a turn. Feedback gives them information.

A student throws a ball

The ball misses.

Another attempt provides more practice.
It may not reveal what to change.

An illustrative teacher cue

“Next time, step forward with your opposite foot.”

The learner gets something specific to try.

Why PE matters: motor competence is associated with physical activity and fitness. School PE is an important place to develop it.

Does feedback help? And does its design matter?

01 Feedback or no feedback?

Some PE studies report improvements.
Others find no significant difference.

02 Which format and content?

“Good effort,” a movement cue, and a video replay all provide different kinds of information.

The gap: the authors identified a need to synthesize intervention evidence specifically from school PE.

One feedback message. Three ways to describe it.

"Good effort. Next time, step forward with your opposite foot."

Who? / Provider

The teacher

A peer can also provide feedback.

How? / Format

Spoken words

Video is another delivery format.

What? / Content

Praise + information

Encouragement plus a movement cue.

A single intervention can combine a provider, a format, and several types of content.

Paper §§2.3, 4.2–4.3 · Message is an illustrative example

What learners need may change with experience and the task.

Cognitive / 认知

How do I do it?

Understand the movement.

Associative / 联结

How do I get consistent?

Refine the movement pattern.

Autonomous / 自动化

What needs adjusting?

Fine-tune a familiar skill.

The task matters, too.

One throw has a clear start and finish. Swimming repeats continuously.
Responding to a moving opponent adds changing perceptual demands.

An explanation to investigate. The review does not establish stage-specific rules or an optimal feedback frequency.

Paper §§2.3, 4.2–4.3 · Fitts & Posner (1967), cited in the paper

The review separates usefulness from the choice of feedback.

Organize

Bring 23 studies together.

Focus on feedback interventions in real school PE classes.

Compare

Separate the design questions.

Presence of feedback, delivery format, and feedback content.

Judge

Make uncertainty visible.

Grade support using study quality and consistency of findings.

The contribution is an evidence synthesis and a research agenda. It does not introduce a new theory of learning.

Paper §§1, 3.4, 5–6

How did studies enter the review?

7 databases searched through April 2021 → **139** potentially relevant studies after screening → **23** included studies 21 + 2 from reference lists

What qualified?

A deliberate feedback intervention · School PE

General elementary or secondary students

Peer-reviewed English full text

Two reviewers independently screened studies. A third helped resolve disagreements. PRISMA guides transparent reporting.

Paper §§2.1–2.2, 3.1 · Simplified selection process

What kind of evidence did they find?

23

studies published
1987–2019

School students. Different skills.

12 elementary · 7 middle · 4 high school studies
Throwing, volleyball, swimming, gymnastics, and more.

Many small studies.

Samples ranged from 3 to 950 students.
16 studies had fewer than 100 participants.

Pre–post: did students improve after practice?

Improvement may reflect practice as well as feedback.

Randomized comparison: did groups differ?

Five studies were classified as RCTs by the authors.

Paper §3.2 & Table 1 · Designs as classified by the authors

A study’s quality and a conclusion’s support are different judgments.

One study / PEDro

How trustworthy is its design?

Random allocation, blinding, reporting, and related criteria.

8 higher quality / **15** lower quality

Authors’ threshold: ≥5 out of 10.

Across studies / Best-evidence synthesis

How much support does a claim have?

Consider quality, quantity, and consistency of findings.

Strong · Moderate · Limited
Conflicting · Insufficient

Consistent: ≥⅔ significant in the same direction.

No pooled effect size: interventions were too varied. “Strong evidence” describes support under this grading scheme, not the size of improvement.

Paper §§2.4–2.5, 3.3 · PEDro & best-evidence synthesis

Feedback has the clearest support. The best format is less certain.

Comparison	Reported findings	Evidence rating
Feedback vs. none	10 of 13 favor feedback	**Strong**
Praise + information vs. praise	2 of 3 favor the combination	**Moderate**
Visual vs. verbal	3 of 3 favor visual	Limited
Information vs. praise or correction	4 of 7 favor information	Conflicting

Counts follow the authors' synthesis. A study can contribute more than one result; these are not all independent replications.

Paper §§3.4, 6 · Authors' evidence classifications

Encouragement can come with something specific to try.

Praise alone

"Good effort."

May support motivation.
Provides little movement information.

Praise + information

"Good effort.
Step forward with your opposite foot."

Moderate evidence favors combining praise and information. The review does not establish the mechanisms behind that benefit.

A useful synthesis. An uneven evidence base.

What strengthens the review

A transparent process

Seven databases and reference searches.

Independent screening and quality ratings.

Study characteristics reported in detail.

What limits the conclusions

Quality and comparability

15 of 23 studies rated lower quality.

No study blinded outcome assessors.

Different tasks and combined interventions.

What remains unresolved?

The average benefit, optimal frequency, long-term retention, and which feedback suits which learner.

Does the improvement remain when the feedback disappears?

During practice

Performance

Can the learner do better with guidance available?

After a delay

Retention

Can they still perform without that guidance?

In another setting

Transfer

Can they use the skill in the real target task?

Presenter’s critique: the review does not systematically separate these outcomes. Durable learning needs more than a better practice score.

Presenter’s evaluation of the review’s outcome synthesis

XR can test feedback designs. It cannot assume they work.

Replay a movement	Compare visual replay with the same spoken guidance. Video evidence does not establish the benefit of 3D overlays.
Pair encouragement with a cue	Give a concrete next action. Praise findings do not directly establish benefits from badges.
Vary when guidance appears	Test frequency or learner-requested feedback. Feedback benefits do not imply feedback on every attempt.

Design hypotheses beyond the paper’s direct evidence. Evaluate tracking accuracy, independent performance, retention, and transfer.

Presenter’s XR design hypotheses, informed by paper §§4–5

An AR throwing coach, with a testable learning claim.

After an attempt

Replay. Encourage. Give one movement cue.

Show the throw alongside an expert example.
“Good effort. Step forward with your opposite foot.”

Proposed prototype, not a tested system in the review.

Compare equal practice time

A. Praise alone

B. Praise + movement information

C. The same information + visual replay

Then test without assistance:
accuracy · delayed retention · real throws

The comparison isolates the added value of information and replay. Improvement in the headset is only the first outcome.

Feedback helps.
Its design still needs testing.

Clearest support	Feedback compared with none.
Moderate support	Praise combined with movement information.
Still open	Best format, frequency, and learner-specific design.

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